

Aim: - To construct and set up a programmable timer with features such as countdown, auto reset, and output alert

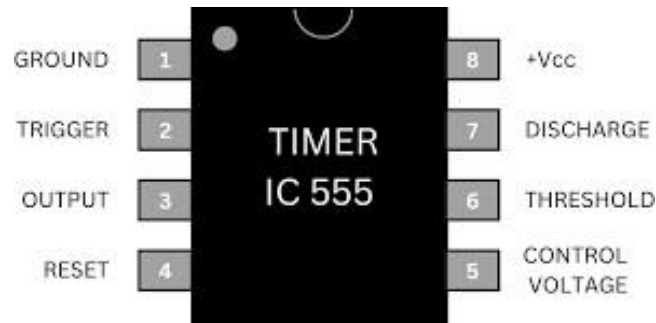
Components required:

Component name	Quantity
NE555P Precision Timer IC- DIP-8 Package	1
CD4511BE - CMOS BCD To 7 Segment Latch Decoder Driver DIP-16 Package	4
CD4510 - Binary Up/Down Counter With Preset Function IC - DIP-16 Package	4
7 Segment Display-Common Cathode - Red - 14mm (0.56 ")	4
4.8mm 360 Degree Super Wide Angle Straw Hat LED - Red	1
Single Sided Universal PCB Prototype Board- 18x30CM Green	1
16 Pin IC Base For - DIP-16 Package	As many as required
74LS04 quad NOR IC DIP-14 Package	1
74LS08 quad AND IC DIP-14 Package	4
6x6x10mm Tactile Push Button Switch Black - 4Pin DIP Package	2
1K,10k and 100k Ohm 1 Watt Metal Film Resistor 1% Tolerance	As many as required
10uF & 100nF 25V Electrolytic Capacitor	As many as required
Soldering machine and wire	-
Battery- 5V	As many as required
DIP Switch	1

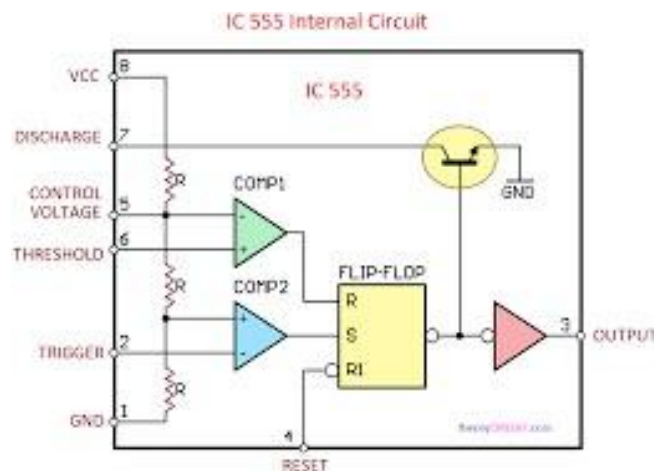
Pin Configurations, pin diagram and functionality of each pin for IC's:

1. NE555P Precision Timer IC:

- Pin diagram:



- Internal circuit diagram:



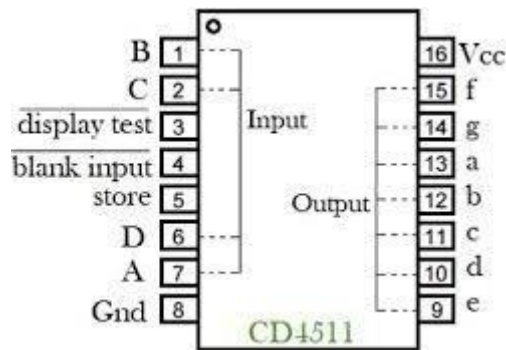
- Functionality of each pin in NE555 timer ic:

Pin No → Pin Name → Function / Description:

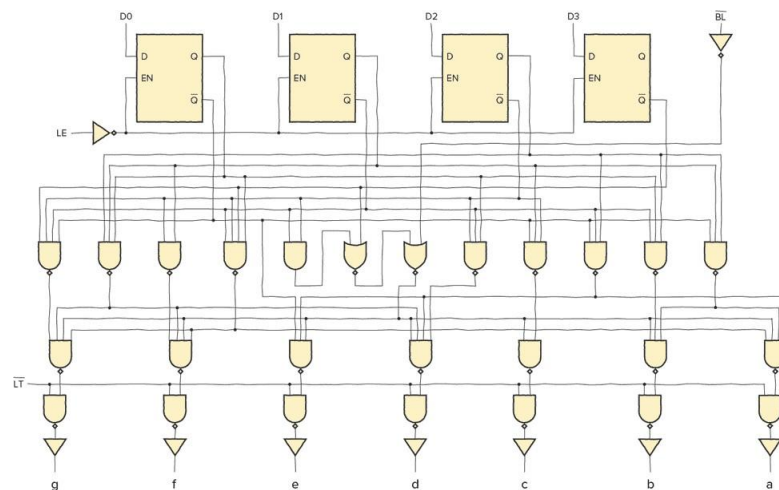
- 1 → GND (Ground) → Connects to circuit ground; reference for all voltages.
- 2 → Trigger → Starts timing when voltage drops below $1/3$ VCC
- 3 → Output → Provides HIGH or LOW output signal up to 200 mA.
- 4 → Reset → Active LOW; resets output, ending timing cycle.
- 5 → Control voltage → Adjusts threshold level; for modulation or noise bypassing.
- 6 → Threshold → Ends timing when capacitor voltage exceeds $2/3$ VCC.
- 7 → Discharge → Discharges timing capacitor via internal transistor.
- 8 → VCC → Power supply input, typically +5 V to +15V.

2. CD4511BE - CMOS BCD To 7 Segment Latch Decoder Driver IC:

- Pin diagram:



- Internal circuit diagram:



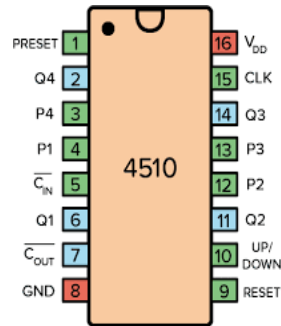
- Functionality of each pin in CD4511BE ic:

Pin No → Pin Name → Function / Description:

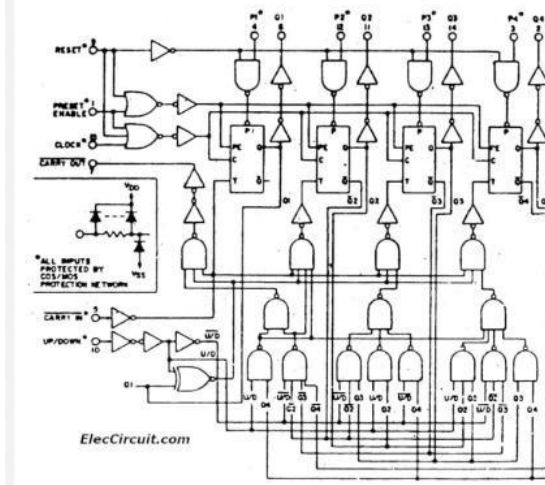
- 1 → B → BCD input bit; second least significant.
- 2 → C → BCD input bit; second most significant.
- 3 → LT (Lamp Test) → Lights all segments when pulled LOW for testing. Display.
- 4 → BI (Blanking Input) → Turns OFF all segments when LOW; active LOW input.
- 5 → LE (Latch Enable) → Latches BCD inputs when HIGH; direct when LOW.
- 6 → D (MSB) → Least significant bit of BCD input data.
- 7 → A (LSB) → Most significant bit of BCD input data.
- 8 → GND → Ground reference pin; connect to 0V.
- 9 → e → Segment 'e' output to 7-segment display.
- 10 → d → Segment 'd' output to 7-segment display.
- 11 → c → Segment 'c' output to 7-segment display.
- 12 → b → Segment 'b' output to 7-segment display.
- 13 → a → Segment 'a' output to 7-segment display.
- 14 → g → Segment 'g' output to 7-segment display.
- 15 → f → Segment 'f' output to 7-segment display.
- 16 → VDD → Positive supply voltage, typically +3V to +15V.

3. CD4510 - Binary Up/Down Counter With Preset Function IC:

- Pin diagram:



- Internal circuit diagram:



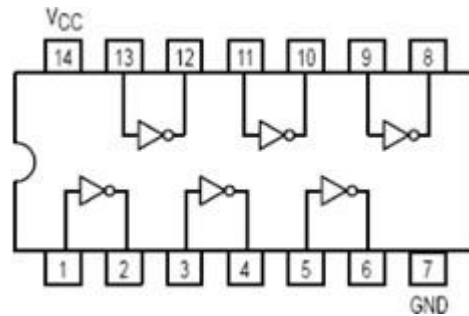
- Functionality of each pin in CD4510 IC:

Pin No → Pin Name → Function / Description:

- 1 → Preset Enable(PE) → Loads input data (A–D) when LOW; count disabled When HIGH
- 2 → Q4 → Output for BCD bit 4(MSB).
- 3 → D(MSB) → Most significant bit of BCD input data.
- 4 → A (LSB) → Least significant bit of BCD input data.
- 5 → Carry In (CI) → Enables counting when HIGH; used for cascading.
- 6 → Q1 → Output for BCD bit 1(LSB).
- 7 → CO (Carry Out) → Output pulse for cascading multiple counters.
- 8 → GND → Ground reference pin; connect to 0V.
- 9 → Borrow Out (BO) → Used in down counting for cascading counters.
- 10 → DOWN/UP → Determines counting direction; HIGH = down, LOW = up.
- 11 → Q2 → Output for BCD bit 2.
- 12 → B → BCD input bit; second least significant.
- 13 → C → BCD input bit; second most significant.
- 14 → Q3 → Output for BCD bit 3.
- 15 → Clock → input for counting pulses; edge-triggered
- 16 → VDD → Positive supply voltage, typically +3V to +15V.

4. 74LS04 Hex Inverter IC :

- Pin diagram along with internal circuit diagram:



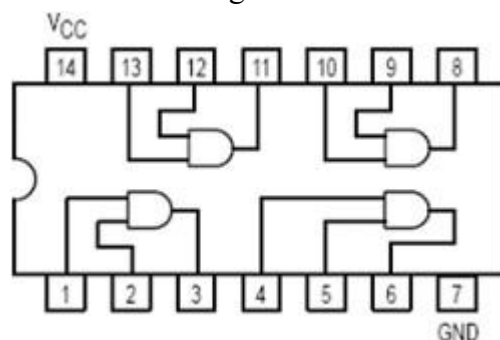
- Functionality of each pin in 74LS04 Hex Inverter IC

Pin No → Pin Name → Function / Description:

- 1 → 1A → Input of inverter 1.
- 2 → 1Y → Output of inverter 1.
- 3 → 2A → Input of inverter 2.
- 4 → 2Y → Output of inverter 2.
- 5 → 3A → Input of inverter 3.
- 6 → 3Y → Output of inverter 3.
- 7 → GND → Ground (0V).
- 8 → 4Y → Output of inverter 4.
- 9 → 4A → Input of inverter 4.
- 10 → 5Y → Output of inverter 5.
- 11 → 5A → Input of inverter 5.
- 12 → 6Y → Output of inverter 6.
- 13 → 6A → Input of inverter 6.
- 14 → VCC → Supply Voltage (+5V).

5. 74LS08 quad AND IC:

- Pin diagram along with internal circuit diagram:



- Functionality of each pin in 74LS04 Hex Inverter IC

Pin No → Pin Name → Function / Description:

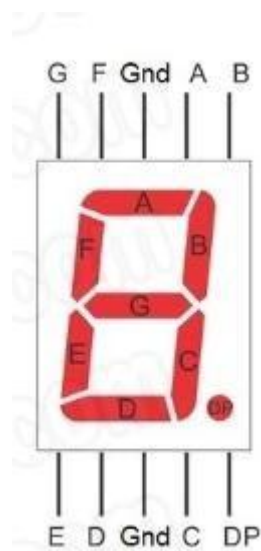
- 1 → 1A → Input A of gate 1.
- 2 → 1B → Input B of gate 1.
- 3 → 1Y → Output of gate 1.
- 4 → 2A → Input A of gate 2.
- 5 → 2B → Input B of gate 2.
- 6 → 2Y → Output of gate 2.

- 7 → GND → Ground (0V).
- 8 → 3Y → Output of gate 3.
- 9 → 3A → Input A of gate 3.
- 10 → 3B → Input B of gate 3.
- 11 → 4Y → Output of gate 4.
- 12 → 4A → Input A of gate 4.
- 13 → 4B → Input B of gate 3.
- 14 → VCC → Supply Voltage (+5V).

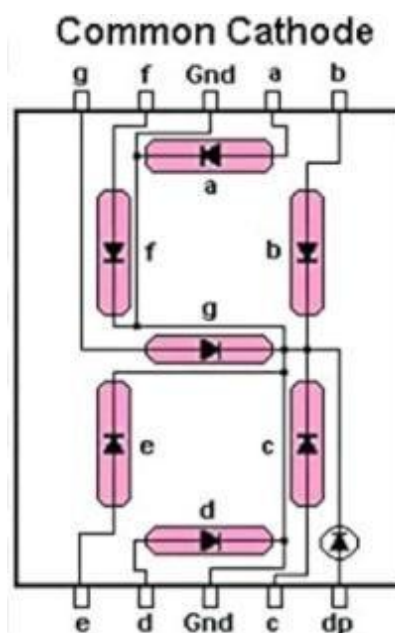
Other components and pin configurations and functionality of each pin:

1. 7 Segment Display-Common Cathode - Red - 14mm (0.56)

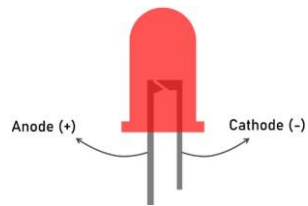
- Pin configuration:



- Internal circuit diagram:



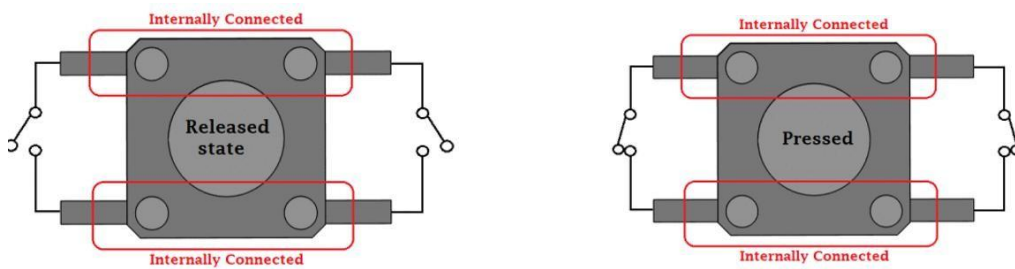
2. LED:



3. IC base:



4. Push buttons:



5. Switch:



Functioning and uses of all components for the project:

1) NE555 timer IC:

i. Functioning of 555 Timer IC:

- Voltage Comparator Action – Compares input voltage with $1/3 V_{CC}$ and $2/3 V_{CC}$ to control the flip-flop inside the IC.
- Flip-Flop Control – The internal SR flip-flop changes state based on the comparators, turning the output HIGH or LOW.
- Discharge Control – Pin 7 discharges the timing capacitor to reset the timing cycle automatically.
- Timing Generation – Using external resistor-capacitor (RC) network, the IC can generate precise time delays, oscillations, or pulse durations.

ii. Usage in the circuit:

- Countdown Timer – Generates a delay corresponding to the countdown period before triggering the output.
- Auto-Reset Function – Discharges the capacitor automatically via pin 7 to restart timing cycles repeatedly.
- Output Alert / Alarm – Drives a buzzer, LED, or other alert devices when the countdown completes.
- Adjustable Timing – Using variable resistors or potentiometers in the RC network, allows programmable timing for different intervals.

iii. 555 Timer Frequency Formula: -

For a stable mode:

$$f = \frac{1.44}{(R1 + 2 * R2) * C}$$

Where:

- $R1 = 1k\Omega$
- $R2 =$ Potentiometer (adjustable, e.g. $10k\Omega - 100k\Omega$)
- $C = 10\mu F$ Assuming $R2 = 100 k\Omega$:

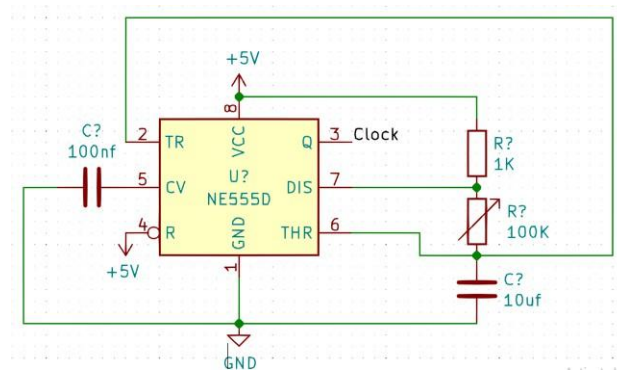
Approx. we get 0.7 Hz (1 pulse every ~1.4 seconds)

So we can adjust the potentiometer to fine-tune the frequency closer to 1 Hz (1 pulse per second), which is ideal for a clock.

Using the 555 timer with our chosen values:

- Produces a square wave pulse close to 1 Hz
- Drives the CD4026 to increment every second
- Allows time adjustment through the potentiometer

iv. Timer circuit diagram(individual):



2) CD4511BE – CMOS BCD To 7 Segment Latch Decoder Driver and 7 segment display:

i. Functioning of CD4511BE IC:

- BCD to 7-Segment Decoder/Driver :Converts a 4-bit binary-coded decimal (BCD) input (0–9) into the corresponding 7-segment display outputs (a–g).
- Latch Enable (LE) Function: When LE = HIGH, the IC latches the input, so the output does not change even if BCD input changes.
- Blanking Input (BI): When BI = LOW, the IC turns off all segments, blanking the display for applications like multiplexing or power saving.
- Lamp Test (LT) Function: When LT = LOW, all segments are turned ON, allowing a quick test of the 7-segment display.

ii. Usage in the circuit:

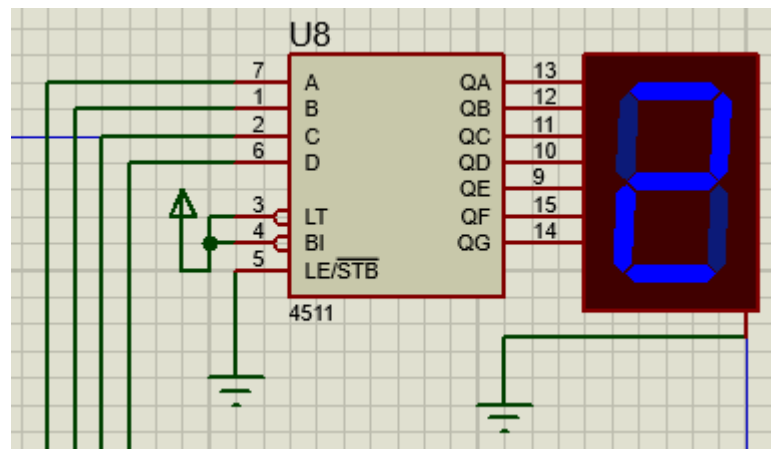
- Display Timer Counts: Shows current count or time remaining in countdown timers on a 7-segment display.
- Visual Feedback for Alerts: Can display numbers or sequences for alert signalling (like 0 when time is up).
- Interface with Digital Logic: Connects to counters, microcontrollers, or timers to drive a display directly without extra decoding circuitry.
- Multiplexed Displays: Supports multiple 7-segment displays with blanking and latching for multi-digit timers.

iii. Conversion table:

- Truth(logic) table for BCD to seven segment display conversion:

Decimal Digit	Input lines				Output lines							Display pattern
	A	B	C	D	a	b	c	d	e	f	g	
0	0	0	0	0	1	1	1	1	1	1	0	0
1	0	0	0	1	0	1	1	0	0	0	0	1
2	0	0	1	0	1	1	0	1	1	0	1	2
3	0	0	1	1	1	1	1	1	0	0	1	3
4	0	1	0	0	0	1	1	0	0	1	1	4
5	0	1	0	1	1	0	1	1	0	1	1	5
6	0	1	1	0	1	0	1	1	1	1	1	6
7	0	1	1	1	1	1	1	0	0	0	0	7
8	1	0	0	0	1	1	1	1	1	1	1	8
9	1	0	0	1	1	1	1	1	0	1	1	9

iv. Circuit diagram(individual):



3) CD4510 - Binary Up/Down Counter With Preset Function IC:

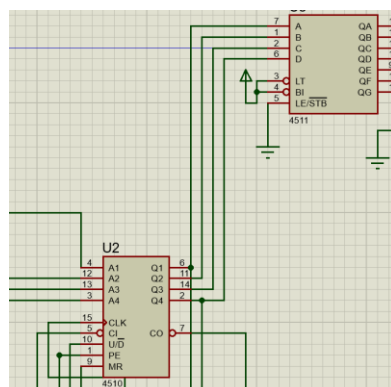
i. Functioning of CD4510 IC:

- Counting: Counts in BCD (0–9) format on each clock pulse applied to the CLK input.
- Up/Down Control: Can increment (up) or decrement (down) depending on the logic at the Up/Down input.
- Presetting: Allows loading a specific number via the parallel inputs (A–D) for starting a countdown.
- Reset/Enable: Can be reset to 0 or disabled via the master reset (MR) and enable (PE) pins, supporting auto-reset functionality.

ii. Usage in the circuit:

- Countdown Timer: Can start from a preset value and decrement to zero to trigger an event.
- Auto-Reset Feature: After reaching zero, the counter can automatically reset for repetitive timing cycles.
- Output Alert: Can be interfaced with LEDs, buzzers, or alarms to indicate when the timer reaches a specific count.
- Digital Display Interface: Works with BCD to 7-segment decoders (like 4511) to show the countdown on a 7-segment display.

iii. Diagram:



4) 74LS02 quad NOR IC :

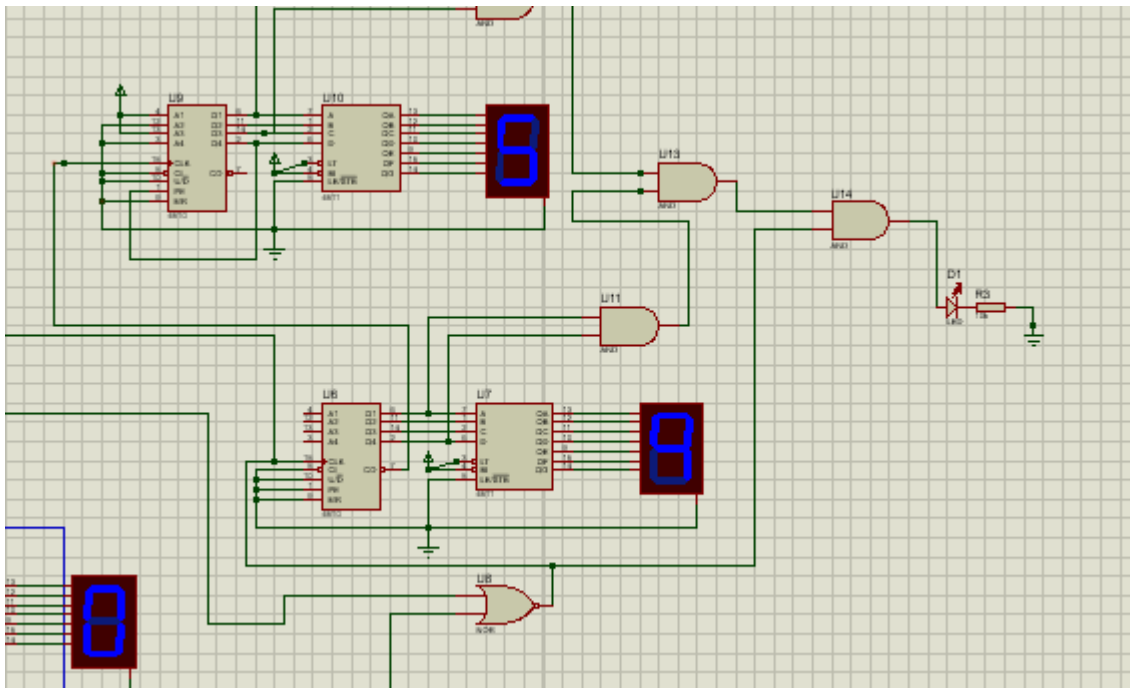
i. Functioning of 7402 IC:

- NOR gate output is visualised when both the inputs are '0'. So both the carry out of the are connected in the form of inputs to the NOR gates when 00 appears on the seconds section of the timer. Both the carry out will become 0 and the output of the NOR gate is connected to the clock pulse of the minutes section and it is also used for output alert.

ii. Usage in the circuit:

- For output alert and updation in the minutes section.

iii. In the circuit:



5) 74LS08 quad AND IC and LED:

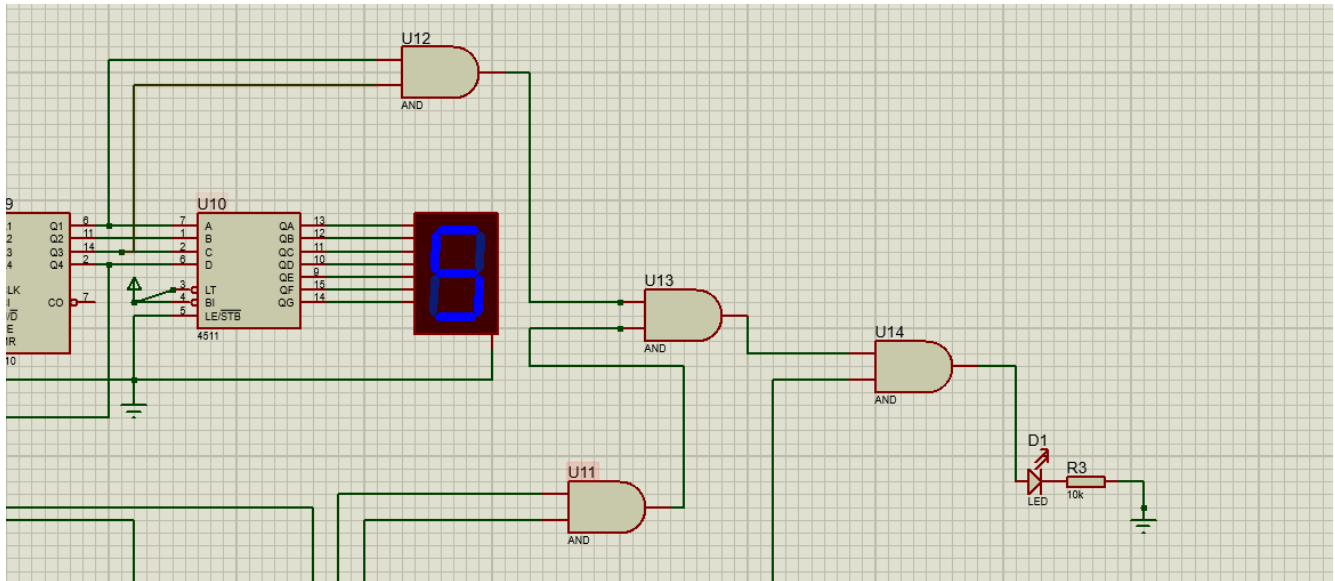
i. Functioning of 74LS08 IC:

- As we know that AND gate yields output when both the inputs are 1.
- Here in the circuit to detect 00:00 from the timer ,(actually it is 59:00) , AND gates are used to detect 5 in ten's digit in minutes and 9 is used in unit's digit of minutes. 5-0101 and 9-1001 for 5 from 2nd MSB and LSB are the inputs of the AND gate and for 9- MSB and LSB are used as inputs of 2nd and gate and another AND gate is used to detect '59' in the minutes section.
- Similarly from the NOR gate (it will detect 00 via the carry out) and from that output it will detect 59:00 (00:00).
- To this output a LED is connected to visualize the output(output alert)

ii. Usage in the circuit:

- For the updation of ten's digit in seconds, minutes and hours in the timer circuit after one cycle of flow.

iii. In the logic circuit:



6) Switches and push buttons:

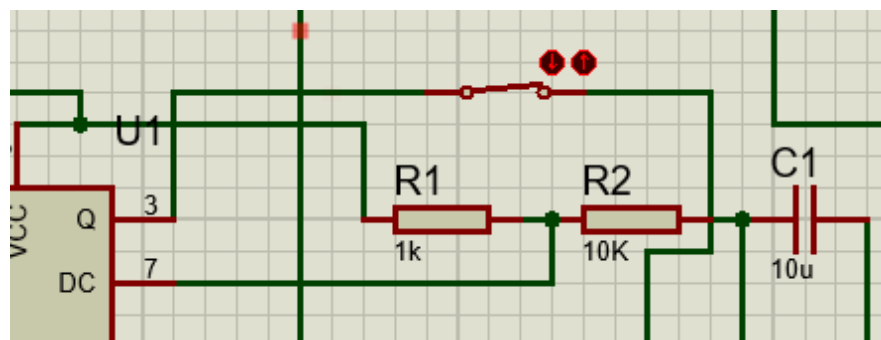
1. Functionality of Pushbuttons and switches:

- A switch is used to stop the flow of count when the switch is opened, count gets stopped and switch is closed, count resumes.
- When the switch is opened if we want to control the count from a particular point then.. by placing a push button we can actually manual reset the timer.

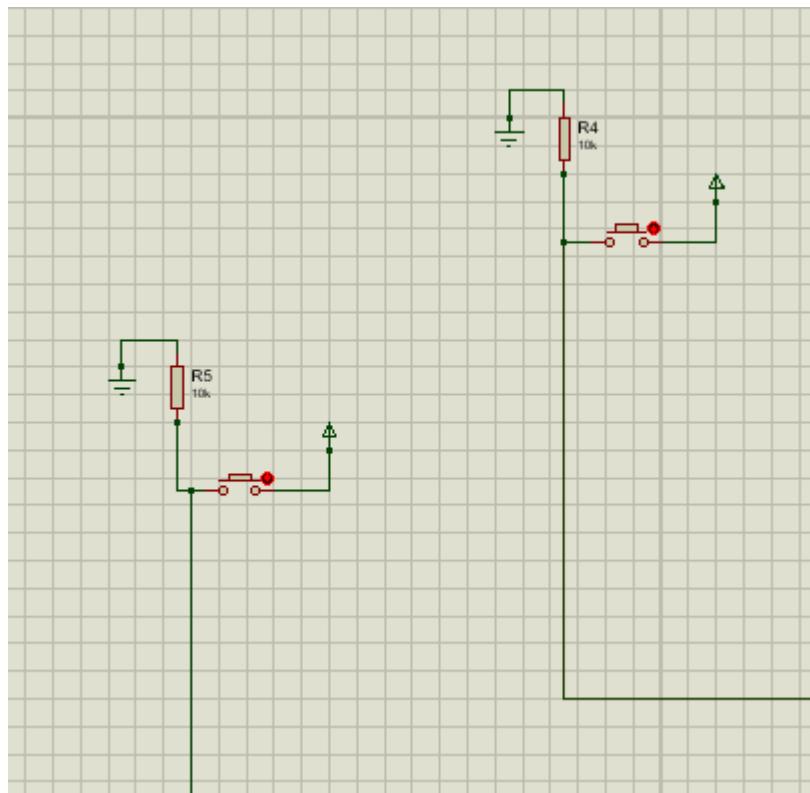
2. Usage in the circuit:

- To stop the flow, we can actually use the switch.
- For manual setting of the time we can use the push buttons.

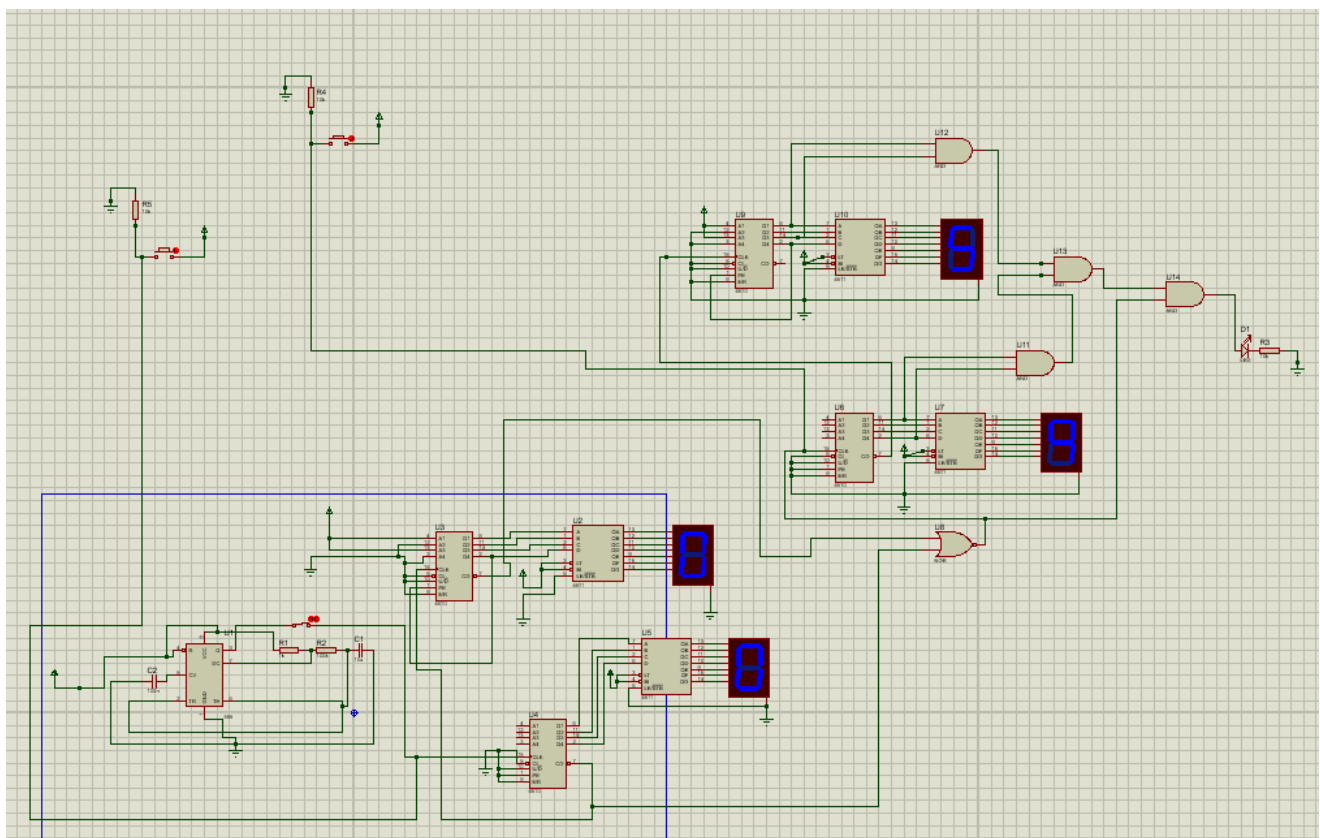
3. Circuit diagram: (placed at the output of the timer(switch))



For the clock pulse generation in the 4510 counter a push button was placed:



Full Circuit Diagram:



Entire working of the circuit:

- **Clock Pulse Generation using NE555 Timer**

The NE555 timer is configured in astable mode with a resistor-capacitor network to produce a ~ 1 Hz pulse. This clock pulse acts as the timing base, generating one pulse per second. The output from pin-3 of the 555 is fed into the CD4510 counter to drive the seconds counting operation.

- **BCD Counting using CD4510 Up/Down Counters**

Each CD4510 IC counts in Binary Coded Decimal (0-9 range). The first counter counts seconds from 0-9 and carries to the next counter after completing 10 cycles. Similarly, minutes and hours follow the same BCD pattern. The counter can also be preset using push buttons to start from a user-defined time.

- **Display Output through CD4511 BCD-to-7 Segment Driver**

The BCD output from each CD4510 counter is fed into a CD4511 decoder IC. CD4511 converts the 4-bit BCD number to signals that light the corresponding segments of a 7-segment display. This shows the current seconds and minutes in real-time as the countdown progresses.

- **Logic Control & Time Detection using NOR and AND Gates**

74LS02 NOR gates detect when both seconds digits become zero (00), providing reset carry logic for minute decrementing. 74LS08 AND gates detect 59 in the minute section, signalling the transition between minutes and hours or ending the countdown. These logic combinations ensure proper rollover from 59 $\rightarrow$ 00 and control alarm triggering.

- **User Control, Reset, and Alert System**

Switches pause or resume the timer by enabling/disabling the clock signal. Push buttons allow manual setting or resetting of time. When the final count reaches 00:00, the logic gates activate an LED/buzzer alert to indicate countdown completion, signalling the timer has finished its cycle.

- **Automatic Carry and Borrow Mechanism**

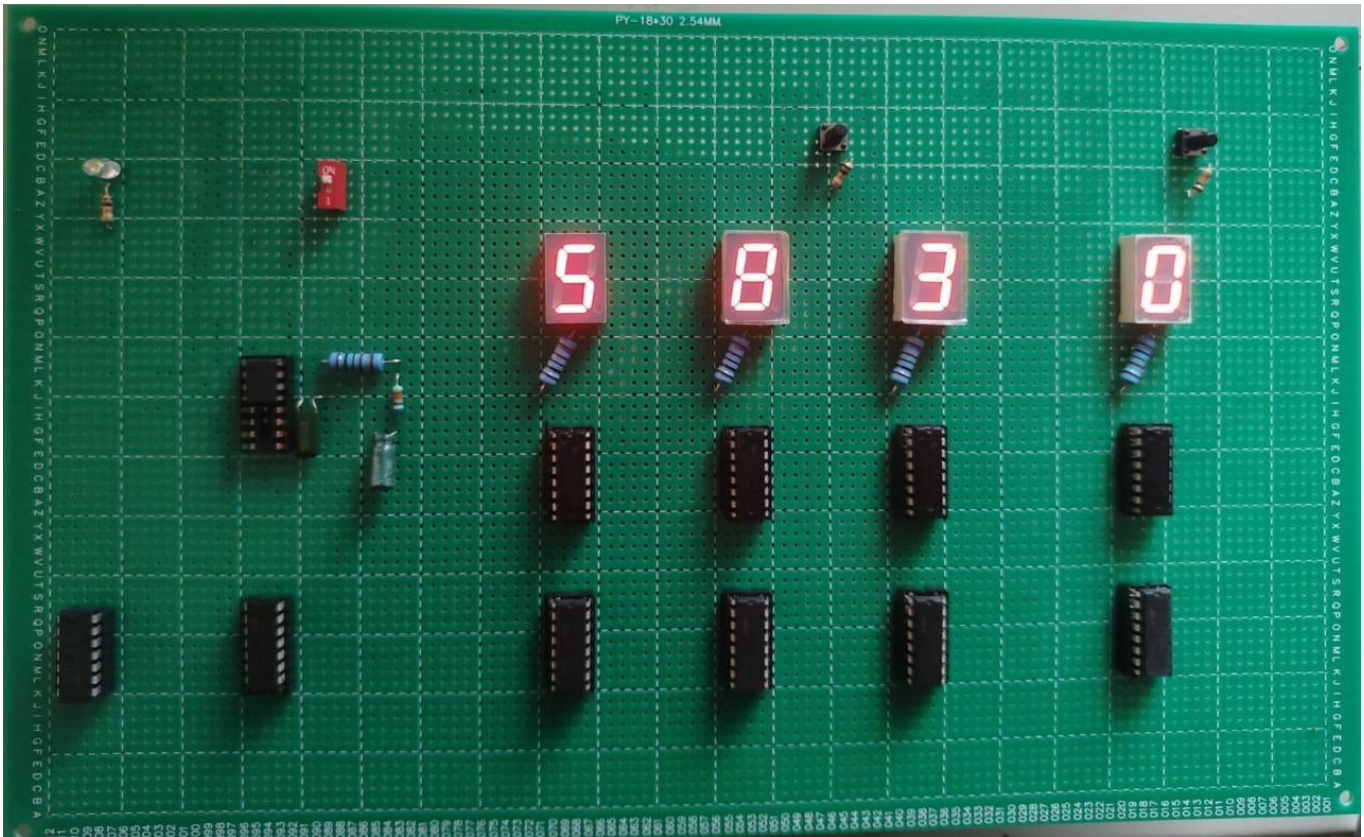
When the seconds counter completes its countdown (reaches 00), a carry/borrow signal is generated and passed to the minutes counter. This ensures automatic decrement of minutes without user intervention, maintaining continuous countdown flow across the time units.

- **Power-On Initialization & Stability**

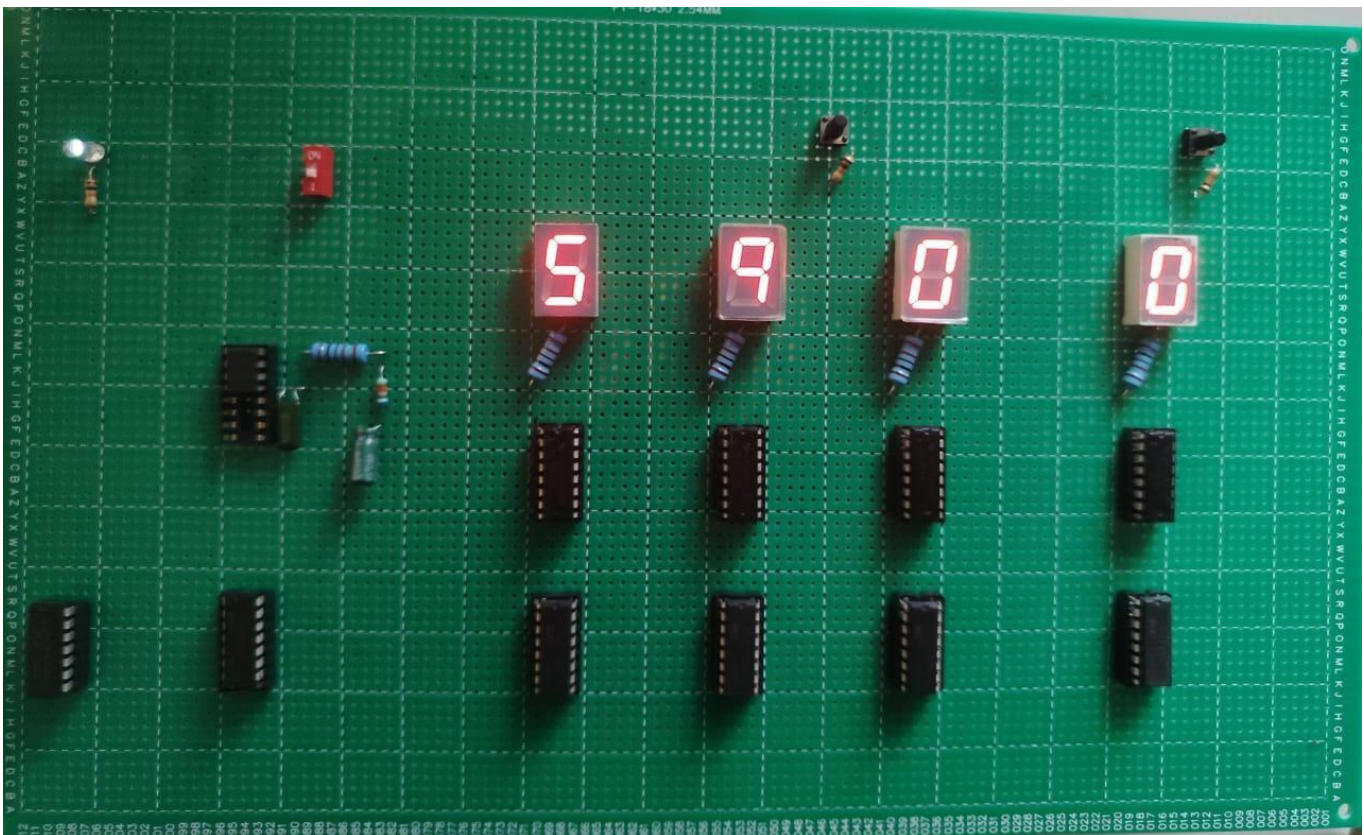
On powering the circuit, reset pins initialize all counters to zero or a preset value. The RC stabilization network and debouncing on switches prevent false triggering. This ensures stable countdown operation and accurate time display without glitches or bounce-errors.

Output Images:

Normal Working of Timer:



Output Alert:



Problems faced during the implementation: -

1. Formation of solder bridges (short circuit) and cold joints.
2. Pads lifting from PCB.

Problems resolved using solutions: -

1. Ensure the soldering iron temperature is appropriate and the solder is applied correctly.
2. Employ high-precision placement equipment and accurate component placement techniques.
3. Proper placement prevents excessive stress on the pads.

Result:

This project successfully demonstrated the construction of digital Timer using IC 4010, IC 4011, along with a common cathode display. The Timer displayed time in MM:SS (1 HOUR) format accurately, with stable timing provided by the 555 IC. The use of PCB made the design compact and durable, ensuring reliable performance.

References:

- NE555 (Timer IC):

<https://www.ti.com/lit/ds/symlink/ne555.pdf> [Texas Instruments+2RS Components+2](#)

<https://www.alldatasheet.com/datasheet-pdf/pdf/17972/PHILIPS/NE555.html>

[AllDatasheet](#)

- CD4510 (Presettable Up/Down Counter):

<https://www.ti.com/lit/gpn/CD4510B> [Texas Instruments+1](#)

<https://www.alldatasheet.com/datasheet-pdf/pdf/26904/TI/CD4510.html>

[AllDatasheet+1](#)

- CD4511B (BCD to 7-Segment Driver):

<https://www.ti.com/lit/gpn/CD4511B> [Texas Instruments](#)

<https://www.alldatasheet.com/datasheet-pdf/pdf/26905/TI/CD4511.html>

[AllDatasheet+1](#)